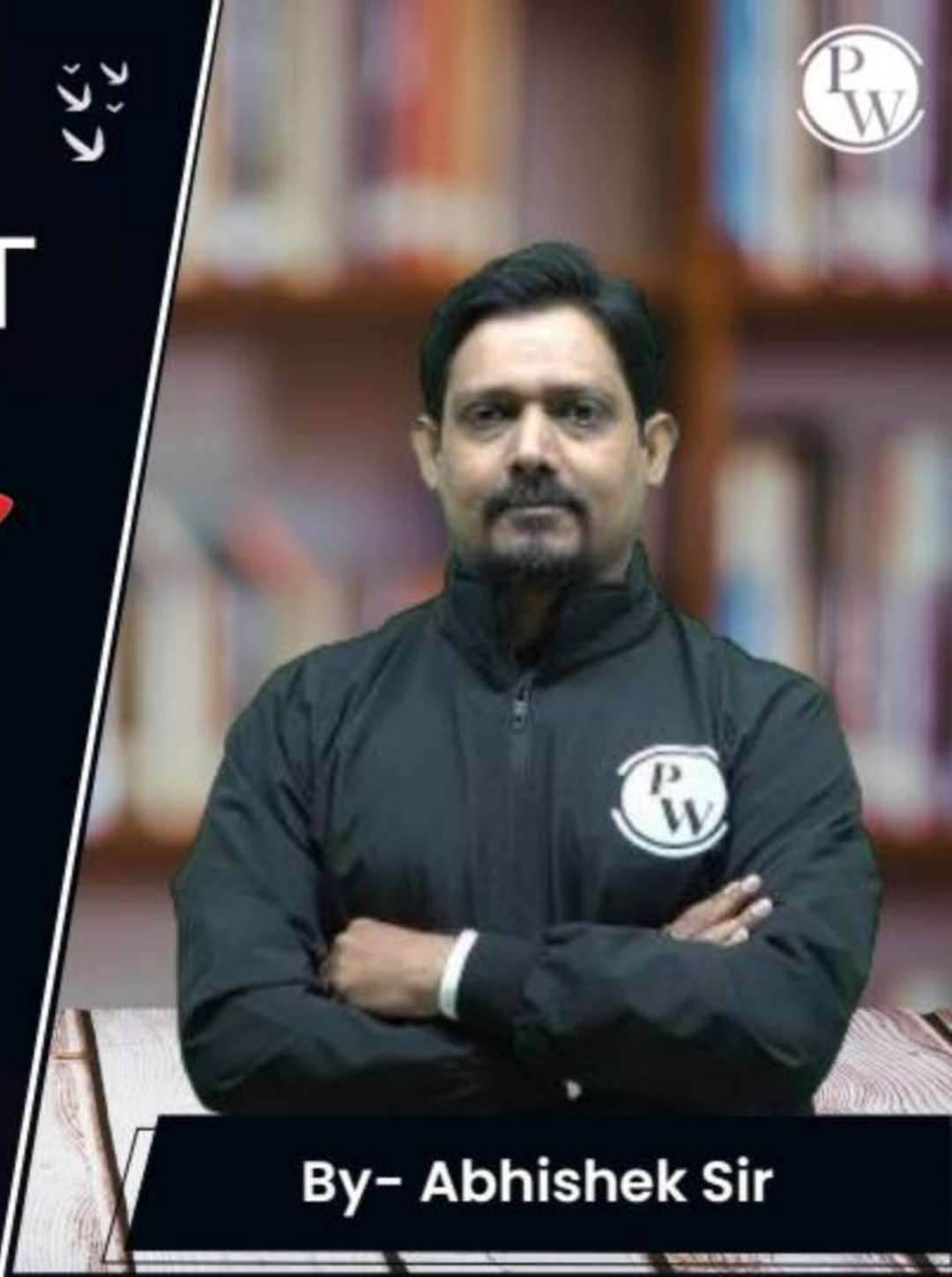
Computer Science & IT



Fundamental of C Language

Function & Storage Class

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



Topic	Recursion
Topic	factorial function
Topic	Fibonacci Series
Topic	
Topic	

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Topics to be Covered



- Topic — Local variable
- Topic — global variable
- Topic — Nested function
- Topic
- Topic

Slide



Local & Global variable

```
#include<stdio.h>
                    a = 20
void fun (int a) {

   int b;

   b = a + 100; // b = 20 + 100 = 120

   printf ("%d", b); // 120

} ↖ function is terminating
```

```
int main() {

   int i = 20,

   fun (i);

   printf ("%d", i);

   return 0;

}
```

Local

Access limited

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Local & Global variable

With every variable two asped are associated

1 Scope : The place where we can accass the variable is called Scope of variable

Access. : update, output

Local variable Scop is within { , }

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Local & Global variable

Local variable scope is within curly brackets.

2. Lifetime of a variable :- Local variable, every time function is in execution then memory for Local variable is allocated. Upon termination of a function memory is deallocated.

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Local & Global variable

Compile time allocation :- If memory is allocated during compilation then its called compile time allocation

Run time allocation :- If memory is allocated during run time the its called as run time allocation.

Local variable allocates memory at runtime

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Local & Global variable

```
#include<stdio.h>
void fun (int a) {
    int b;
    b = a+100; // b = 20+100=120
    printf ("%d", b); //120
}  ↖ function is terminating
```

```
int main(){
    int a=40;
    fun(a);
    printf("%d", a);
    return 0;
}
```

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Local & Global variable

In different scope two Local variable

can Have same name.

But Not in same scope



Local & Global variable




```
#include<stdio.h>
int a=10;
void f1(){
    a=a+20;
}
void f2(){
    a=a+40;
}
```

No declaration of 'a'

```
int main(){
    a=a+40; // a=50
    printf("%d", a); // 50
    f1();
    printf("%d", a); // 70
    f2();
    printf("%d", a); // 110
    return 0;
}
```

50

Slide



Local & Global variable

Global variable :- Global variable can be accessed anywhere in the program. (scope)

global variable lifetime is until the program is running

Global variable allocation done at compile time

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Local & Global variable

Variable a is Not declared but we are accessing it

In c programming it comes out of { } curly brackets to check if its declared globaly

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Local & Global variable



```
#include<stdio.h>
int a=10;
void f1(){
    int a=20;
    a=a+20;
}
void f2(){
    a=a+40;
}
```

a_L | 40 |

if conflict Happen between local & global Local will be preferred.

```
int main(){
    a=a+40;
    printf("%d",a); //50
    f1();
    printf("%d",a); //50
    f2();
    printf("%d",a); //90
    return 0;
}
```

a_G | 10 | 50 | 90 |



Local & Global variable

```
#include<stdio.h>
int a=10;
void f1(){
    int a=20;
    a=a+20;
}
void f2(){
    a=a+40;
}
```

global variable

function Scope by default is global

```
int main(){
    a=a+40;
    printf("%d",a);
    f1()
    printf("%d",a);
    f2();
    printf("%d",a);
    return 0;
}
```

Slide



Nested function

Nested function is function defined within the function

```
#include<stdio.h>
int main() {
    int a = 10;
    void fun() {
        printf("within fun");
    }
    printf("%d", a);
    return 0;
}
```

Slide



Nested function

Nested function is function defined within the function

```
#include<stdio.h>
int main() {
    int a = 10;
    void fun() {
        printf("within fun");
    }
    printf("%d", a);
    fun();
    return 0;
}
```

Slide



2 mins Summary

Topic

Local variable Scope & lifetime

Topic

global variable Scope & Lifetime

Topic

Nested function

Topic

Topic

Slide

THANK - YOU

